1. Apply all the CPU scheduling algorithms for insem examination of OSSP

	AT	BT	Priority
P_1	0	4	3
P_2	0	9	1
P_3	0	6	2
P_4	0	4	3
P_5	0	9	1
P_6	0	6	2
P_7	1	4	3
P_8	1	9	1
P_9	1	6	2

2. Define following terms
* Starvation
* Convoy effect
* Contest Switching
* Aging

3. A Mobile has the following tasks, draw a table with

Task	AT	BT
Music Player	0	20
Touch even	3	1
Whatsapp	5	2
Camera Launch	6	4

(i) Which Scheduling algorithm is best suitable for above problem

(ii) Draw the grann chart for SRTF

(iii) Why selected algorithm for improve user experience

(iv) Which task recieves the fastest response

4. Solve the following problem with round robin algorithm time quantum of R & context with time of 0.5 Draw a table

Process	AT	BT
P_1	0	5
P_2	0	4
P_3	0	2

Solutions

1. FCFS - First Come First Serve

Ready Queue	P1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9
Arrival time	0	0	0	0	0	0	1	1	1

Gantt Chart

P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9

0 4 13 19 23 32 38 42 51 57

TAT = CT - AT

WT = TAT - BT

= CT - AT - BT

P_1
P_2
P_3
P_4
P_5
P_6
P_7
P_8
P_9

Avg TAT
Avg WT

SJN (Short
first (

Process
P_1
P_2
P_3
P_4
P_5
P_6

Smallest B
Gantt Cha

P_1	F

0 4

	AT	BT	CT	TAT	WT
P_1	0	4	4	4	0
P_2	0	9	13	13	4
P_3	0	6	19	19	13
P_4	0	4	23	22	19
P_5	0	9	32	32	23
P_6	0	6	38	38	32
P_7	1	4	42	41	37
P_8	1	9	51	50	41
P_9	1	6	57	56	50
				276	219

Avg TAT $= \frac{276}{9} = 30.667$

Avg WT $= \frac{219}{9} = 24.333$

SJN (shortest Job Next) / shortest Job first (SJF) :-

Process	AT	BT	Process	AT	BT
P_1	0	4	P_7	1	4
P_2	0	9	P_8	1	9
P_3	0	6	P_9	1	6
P_4	0	4			
P_5	0	9			
P_6	0	6			

Smallest BT = 4

Gantt Chart

P_1	P_4	P_7	P_3	P_6	P_9	P_2	P_5	P_8

0 4 8 12 18 24 30 39 48 57

	AT	BT	CT	TAT	WT
P_1	0	4	4	4	0
P_4	0	4	8	8	4
P_7	1	4	12	11	7
P_3	0	6	18	18	12
P_6	0	6	24	24	18
P_9	1	6	30	29	23
P_2	0	9	39	39	30
P_5	0	9	48	48	39
P_8	1	9	57	56	47
				237	180

Avg TAT $= \frac{237}{9} = 26.33$

Avg WT $= \frac{180}{9} = 20$

SRTF (shortest Remaining Time first)

At Time 0 P_1 starts RBT = $4 - 1 = 3$

at time 1 P_1, P_8, P_9 arrive.

P_1	3
P_4	4
P_7	4
P_3	6
P_6	6
P_9	6
P_2	9
P_5	9
P_8	9

Gantt

Priority

Higher

$P_2 \rightarrow$

$0 \rightarrow 9$

$9 \rightarrow 18$

$18 \rightarrow 27$

Priority

$27 \rightarrow 3$

$33 \rightarrow 39$

$39 \rightarrow 45$

Gantt Ch

	CT
P_1	49
P_2	9
P_3	33
P_4	53
P_5	18
P_6	39
P_7	57
P_8	27
P_9	45

Gantt Chart

P1	P4	P7	P3	P6	P9	P2	P5	P8

0 4 8 12 18 24 30 39 48 57

Same values as SJN.

Priority Scheduling - (Non Preemptive)

Highest Priority = 1

$P_2 \rightarrow P_5$

$0 \rightarrow 9 : P_2$

$9 \rightarrow 18 : P_5$

$18 \rightarrow 27 : P_8$

Priority 2

$27 \rightarrow 33 : P_3$

$33 \rightarrow 39 : P_6$

$39 \rightarrow 45 : P_9$

Priority 3:-

$45 \rightarrow 49 : P_1$

$49 \rightarrow 53 : P_4$

$53 \rightarrow 57 : P_7$

Gantt Chart

P2	P5	P8	P3	P6	P9	P1	P4	P7

0 9 18 27 33 39 45 49 53 57

	CT	TAT	WT
P_1	49	49	45
P_2	9	9	0
P_3	33	33	27
P_4	53	53	49
P_5	18	18	9
P_6	39	39	33
P_7	57	56	52
P_8	27	26	17
P_9	45	44	38
		327	270

Avg TAT $= \frac{327}{9}$

$= 36.33$

Avg WT $= \frac{270}{9}$

$= 30$

Priority Scheduling (Preemptive)

At time 0, P_2 has Priority 1

0 → 1 : P_2

0 → 9 : P_2
9 → 18 : P_5
18 → 27 : P_8.

Priority 2 :

27 → 33 : P_3
33 → 39 : P_6
39 → 45 : P_9

Priority 3 :-

45 → 49 : P_1
49 → 53 : P_4
53 → 57 : P_7

So, for this particular data
Preemptive = Non Preemptive

2. (i) Starvation :-

Occurs when a process waits indefinitely or for an excessively long time because other processes are repeatedly selected for execution. Starvation can occur in both preemptive and Non-preemptive scheduling, depending on the algorithm.

(ii) Convoy effect :-

is an important problem associated with non-preemptive scheduling especially FCFS. The long process act like a convoy, causing many short processes to follow behind it.

(iii) Cont...
is
of the
loading
process
between

(iv) Agin...

prevent
increasin...
that h...

3.
(i) SRT...
with th...

(ii) At t...
Musis

RBT =

At tim...

Mu...
To...

0 3
| Music | T... |

(iii) <u>Context</u> <u>Switching</u> :-

is the process of saving the state of the currently running process and loading the saved state of another process so that the CPU can switch between them.

(iv) Aging :-

is a technique used to prevent starvation by gradually increasing the priority of a process that has been waiting for a long time

3.

(i) SRTF always selects the process with the shortest remaing Burst time

(ii) At time 0 : only Music Player.

Musis player has executed for 3 units

RBT = 20 - 3 = 17

At time 3 : Touch Event arrives

	RBT
Musis player	17
Touch Event	1

0	3	4	5	7	11	27
Music	TE	Music	Whatsapp	Camera	Music	

(iii) SRTF improves the user experience because it gives short interactive tasks priority over long CPU-intensive tasks

(iv) RT = First CPU start time − AT

	AT	FS	RT
Music	0	0	0
Touch	3	3	0
Whatsapp	5	5	0
Camera	6	7	1

The fastest Response time are:
Music Player, Touch Event, Whatsapp Notification.

4.

P_1	P_2	$P3$
0	0	0

* P_1 requires 5ms

$P_1 : 0 \rightarrow 2$

Remaining : $5 - 2 = 3$ms

Context Switch = $2 \rightarrow 2.5$

$P_2 \rightarrow P_3 \rightarrow P_1$

* P_2 requires 4

$P_2 : 2.5 \rightarrow 4.5$

remaining $4 - 2 = 2$

$P_3 \rightarrow P_1 \rightarrow P_2$

Context S

* P3 req

P_3

$P_1 \rightarrow$

* P_1 has

P_1 :

$3 - 2 =$

$9.5 \rightarrow 10$

$P_2 \rightarrow P_3$

P_1	CS	P_2
0	2	2.5

P_3	CS	P_1
13.5	14	15

P_1

P_2

P_3

TAT = CT

	TAT
P_1	15
P_2	12
P_3	13.5

Avg waiting

Avg TAT =

Content Switch : 4.5 → 5

* P3 requires 3ms

$P_3 : 5 \to 7$

$3 - 2 = 1$

$P_1 \to P_2 \to P_3$

* P_1 has 3 ms remaining

$P_1 : 7.5 \to 9.5$

$3 - 2 = 1$

$9.5 \to 10$

$P_2 \to P_3 \to P_1$

0–2	2–2.5	2.5–4.5	4.5–5	5–7	7–7.5	7.5–9.5	9.5–10	10–12	12–12.5
P_1	CS	P_2	CS	P_3	CS	P_1	CS	P_2	CS

12.5–13.5	13.5–14	14–15
P_3	CS	P_1

	CT
P_1	15
P_2	12
P_3	13.5

TAT = CT − AT

	TAT	WT
P_1	15	10
P_2	12	8
P_3	13.5	10.5

Avg waiting time = 9.5

Avg TAT = $\frac{15 + 12 + 13.5}{3} = \frac{40.5}{3}$